

HAMEENLINNA LAMMI PAPPILA, FinlandWMO: **027670**Lat: **61.0541N**Lon: **25.0385E**Elev: **410**StdP: **14.48**Time Zone: **2.00 (EUE)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-5.1	-15.8	2.4	-11.4	-8.7	3.5	-4.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	80.7	65.7	77.2	63.4	73.9	62.2	68.4	76.6	66.3	74.5	64.0	71.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	95.5	72.0	63.1	87.8	70.0	61.1	81.5	67.1	32.9	76.8	31.2	74.5	29.5	71.4	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-15.1	84.1	7.9	4.4	-20.8	87.3	-25.5	89.9	-29.9	92.4	-35.7	95.6
			WB	-14.9	70.2	7.7	3.4	-20.5	72.7	-25.0	74.7	-29.3	76.6	-34.9	79.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	18.7	22.1	27.9	38.7	50.3	56.8	62.2	59.3	51.1	40.4	33.5	26.2
	DBStd	16.60	11.94	11.53	8.36	6.45	7.52	6.24	5.95	5.38	5.95	6.46	7.50	10.11
	HDD50	4477	970	781	686	344	89	12	0	3	59	301	495	737
	HDD65	8932	1435	1201	1151	790	456	256	129	190	416	761	945	1202
	CDD50	1087	0	0	0	5	100	216	378	291	92	5	0	0
	CDD65	67	0	0	0	0	2	10	42	13	0	0	0	0
	CDH74	693	0	0	0	0	55	130	395	113	0	0	0	0
	CDH80	113	0	0	0	0	3	14	75	21	0	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	24.3	1.9	1.4	1.3	1.1	1.6	2.7	3.1	2.8	2.0	2.4	2.2	1.8
(15) Precipitation	PrecMax	31.5	3.4	3.1	2.3	2.3	3.1	4.4	8.0	5.4	5.2	6.3	4.5	3.6
	PrecMin	20.4	0.4	0.1	0.2	0.2	0.3	1.0	0.2	0.6	0.7	0.7	0.1	0.4
	PrecStd	2.8	0.7	0.7	0.6	0.6	0.8	1.0	1.7	1.3	1.1	1.3	1.0	0.7
	PrecStd	2.8	0.7	0.7	0.6	0.6	0.8	1.0	1.7	1.3	1.1	1.3	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	41.3	49.8	67.5	79.4	82.1	85.4	82.7	69.8	57.0	49.5	46.2
		MCWB	36.9	36.9	40.8	51.5	59.1	65.4	68.0	67.1	61.0	53.3	48.0	45.3
	2%	DB	36.4	37.5	44.1	60.7	74.9	77.7	82.3	76.6	66.0	53.6	47.6	41.3
		MCWB	35.7	35.9	38.0	47.1	58.2	62.0	67.1	64.2	59.1	50.5	46.6	40.2
	5%	DB	34.5	35.6	40.5	55.7	70.3	74.1	78.7	73.3	62.9	51.9	44.8	38.7
		MCWB	34.0	34.6	36.3	45.2	56.3	61.1	65.8	62.7	57.2	50.1	43.9	37.7
	10%	DB	32.8	34.1	37.6	50.8	66.0	70.2	75.1	69.9	60.3	49.9	42.4	36.5
		MCWB	32.4	33.4	34.6	42.2	53.9	59.0	64.5	61.2	56.4	48.2	41.7	35.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	37.7	42.1	52.5	63.0	68.6	71.0	70.2	62.5	54.7	48.6	45.5
		MCDB	38.0	39.2	47.0	63.7	72.7	75.6	79.6	79.0	67.1	56.4	49.4	46.4
	2%	WB	35.8	36.2	39.0	49.2	60.2	64.8	69.1	66.5	60.5	52.2	46.6	40.5
		MCDB	36.2	37.4	43.3	59.3	69.6	73.6	78.1	72.9	64.3	53.1	47.7	41.2
	5%	WB	34.1	34.9	37.0	45.9	58.1	62.5	67.6	64.2	58.6	50.6	44.0	37.8
		MCDB	34.5	35.7	39.6	53.3	67.6	71.7	76.1	71.1	61.8	51.5	44.6	38.5
	10%	WB	32.5	33.5	35.2	43.5	55.3	60.3	65.7	62.4	56.8	48.3	42.0	35.9
		MCDB	32.9	34.1	37.3	49.8	64.5	68.5	73.8	67.7	59.5	49.6	42.5	36.4
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.8	13.4	16.9	20.2	18.1	17.7	17.1	13.6	10.0	6.4	7.1
		MCDBR	6.0	7.8	15.5	26.5	28.6	25.3	23.1	22.7	17.8	12.0	8.0	8.0
		MCWBR	5.8	7.0	10.5	14.8	14.0	12.4	10.6	11.9	11.5	9.3	7.8	7.8
	5% WB	MCDBR	6.0	7.4	13.0	22.7	23.9	21.4	20.9	19.2	15.8	10.7	7.7	7.7
		MCWBR	5.9	6.8	9.6	13.7	12.8	11.3	10.2	10.8	10.8	9.3	7.6	7.5
(40) Clear-Sky Solar Irradiance	taub		0.257	0.298	0.324	0.357	0.354	0.351	0.373	0.362	0.337	0.322	0.282	0.368
	taud		2.095	2.166	2.241	2.365	2.430	2.462	2.415	2.442	2.507	2.407	2.191	2.617
	Ebn at Noon		147	213	250	266	276	279	268	262	249	207	137	100
	Edh at Noon		13	22	29	31	32	31	32	29	23	18	12	8
(44) All-Sky Solar Radiation	RadAvg		75	254	722	1189	1605	1676	1623	1255	746	321	88	42
	RadStd		14	56	92	124	177	132	158	140	99	56	17	6

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (60 neighbors)		+0.86	N/A	N/A	N/A	N/A	N/A	-261	N/A	N/A	N/A	

Nomenclature: See separate page